

EcoWatt Electricity Savings Tips Extraction Report

Introduction:

We extracted **53 distinct electricity-saving tips** exclusively from three provided sources: the IEA “Saving Energy” webpage, the DOE Energy Saver Guide, and the ResearchGate paper “Efficient Energy Management: Practical Tips for Household Electricity Conservation.”

Each tip was extracted, normalized using Artificial Intelligence (AI) to ensure clear and actionable language, and assigned to a specific category.

A detailed profiling of our methodology, source contributions, and AI-driven preprocessing decisions is presented below.

Methodology

- **Source Extraction & Contribution:** We scoured the three provided sources for actionable electricity-saving tips. We successfully extracted a total of 53 unique tips distributed as follows:
 - **IEA Website (15 tips):** 3 Cooling, 3 Lighting, 5 Behavior (noting that behavioral tips often overlap with other technical categories), 3 Insulation, 1 Water Heating.
 - **Energy Saver Guide (17 tips):** 6 Cooling, 7 Lighting, 4 Electronics.
 - **Efficient Energy Management Practical Tips for Household Electricity Conservation (21 tips):** 1 Electronics, 1 Insulation, 5 Refrigeration, 4 Water Heating, 5 Cooking, 5 Laundry.
- **Categorization Basis:** The categorization process utilizes a hybrid approach. It is important to note that **some sources were already pre-categorized** by their authors (e.g., the Guide contains explicit sections for "Lighting" and "Cooling"). We used these pre-existing classifications as a baseline. For unclassified tips, we manually categorized them based on the primary end-use of the electricity (e.g., tips involving ovens were mapped to "Cooking").
- **AI Normalization & Paraphrasing:** Artificial Intelligence was utilized to modify and refine the extracted tips. The AI translated lengthy, descriptive paragraphs into clear, understandable, and unified language. The AI normalized the data by stripping away overly technical jargon, removing brand-specific terms, and rewriting the advice into concise, second-person imperative statements (e.g., changing "Consumers should consider replacing..." to "Replace...").

- **Deduplication:** Where multiple sources provided similar advice, the AI merged them into a single, comprehensive statement within the appropriate category to avoid redundancy.

Categories and Definitions

- **Cooling (9 tips):** Air conditioning and fan strategies (thermostat settings, fan use).
- **Lighting (10 tips):** Lighting efficiency (LED bulbs, daylighting, turning off lights).
- **Behavior (5 tips):** General energy-saving habits and real-time feedback actions.
- **Insulation (4 tips):** Building envelope (sealing gaps, window treatments).
- **Water Heating (5 tips):** Electric water heating (lowering temps, insulating pipes).
- **Electronics (5 tips):** Consumer electronics (computers, TVs, advanced power strips).
- **Refrigeration (5 tips):** Refrigerator/freezer specific tips (optimal temperatures, seals).
- **Cooking (5 tips):** Efficient use of kitchen appliances (air fryers, matching pan sizes).
- **Laundry (5 tips):** Washing and drying efficiency (cold water, lint trap cleaning).

Source	Tips Extracted	Categories Covered	Comments
IEA Saving Energy Tips	15	Cooling (3), Lighting (3), Behavior (5), Insulation (3), Water Heating (1)	Behavior tips are foundational and often repeat/apply across other equipment categories.
Energy Saver Guide	17	Cooling (6), Lighting (7), Electronics (4)	This source was already heavily pre-categorized, which guided our taxonomy.
Efficient Energy Management Practical Tips for Household Electricity Conservation	21	Electronics (1), Insulation (1), Refrigeration (5), Water Heating (4), Cooking (5), Laundry (5)	Provided the most detailed appliance-specific tips.
Total Unique Tips	53	9 Categories	Deduplicated and AI-normalized.

References

1. **International Energy Agency (IEA).** (n.d.). *Saving Energy*. Retrieved from <https://www.iea.org/topics/saving-energy>
2. **U.S. Department of Energy (DOE).** (2022). *Energy Saver Guide: Tips on Saving Money and Energy in Your Home*. Retrieved from <https://www.energy.gov/sites/default/files/2022-08/energy-saver-guide-2022.pdf>
3. **Mrudula, G. V., Rohit, G., Harshavardhan, G., Dhanush, K., & Anudeep, B.** (2024). *Efficient Energy Management: Practical Tips for Household Electricity Conservation*. *Journal of Modern Technology (JMT)*, 1(1), 1-8.
4. **Google.** (2024). *Gemini* (Large Language Model). [AI tool used for data extraction, normalization, deduplication, and natural language processing of the energy-saving tips]. Retrieved from <https://gemini.google.com>